11. $\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2}$

$$\lim_{x \to 2} \frac{\cancel{(x-2)}(x+3)}{\cancel{x-2}} \quad \lim_{x \to 2} x+3 = 5$$

$S = \underline{-3} + \underline{2} = -1$

$P = \underline{-3} \cdot \underline{2} = -6$

$a = 1$

$b = 1$

$c = -6$

$$\Delta = b^2 - 4ac$$
$$= 1^2 - 4 \cdot 1 (-6)$$
$$= 1 + 24$$
$$= 25$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$x' = \frac{-1+5}{2} = \frac{4}{2} = 2 \quad (x-2)$$

$$x'' = \frac{-1-5}{2} = \frac{-6}{2} = -3 \quad (x+3)$$

$$\lim_{x \to 2} \frac{x^2 - x + 6}{x - 2}$$ não há solução

$a = 1$

$b = -1$

$c = 6$

$S __ + __ = 1$

$P __ \cdot __ = 6$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{\overbrace{-1^2 - 4 \cdot 1 \cdot 6}^{-24}}}{2 \cdot 1} \rightarrow \frac{1 + \sqrt{1 - 24}}{2}$$ não possui raízes reais.

$$\lim_{t \to -3} \frac{t^2 - 9}{2t^2 + 7t + 3}$$

$$\lim_{t \to -3} \frac{t^2 - 9}{2t^2 + 7t + 3} = \lim_{t \to -3} \frac{\cancel{(t-3)}(t+3)}{(t - \frac{1}{2})\cancel{(t-3)}}$$

$$\lim_{t \to -3} \frac{t+3}{t - \frac{1}{2}} \Rightarrow \lim_{t \to -3} \frac{\overset{0}{\cancel{-3+3}}}{-3 - \frac{1}{2}} = 0$$

$a = 2$

$b = 7$

$c = 3$

$$\Delta = b^2 - 4ac$$
$$= 7^2 - 4 \cdot 2 \cdot 3$$
$$= 49 - 24$$
$$= 25$$

$$t = \frac{-b \pm \sqrt{\Delta}}{2 \cdot a}$$

$$t' = \frac{-7 + 5}{4} = \frac{-2}{4} = -\frac{1}{2}$$

$$t'' = \frac{-7 - 5}{4} = \frac{-12}{4} = -3$$

17. $\lim_{h \to 0} \frac{(-5+h)^2 - 25}{h}$

$\lim_{h \to 0} \frac{\cancel{25} - 10h + h^2 - \cancel{25}}{h}$

$\lim_{h \to 0} \frac{h^2 - 10h}{h} =$

$a^2 - 2ab + b^2$

$(-5)^2 - 2(-5h) + h^2$

$25 + 10h + h^2$

$\lim_{h \to 0} \frac{\cancel{h}(-10+h)}{\cancel{h}}$

$\lim_{h \to 0} -10 + h = -10$

19. $\lim_{x \to -2} \dfrac{x+2}{x^3+8}$

$$\lim_{x \to -2} \frac{\cancel{x+2}^{\,1}}{\cancel{(x+2)}(x^2-2x+4)}$$

x^3+8

x^3+2^3

$(a+b)(a^2-ab+b^2)$

$(x+2)(x^2-x2+2^2)$

$(x+2)(x^2-2x+4)$

$$\lim_{x \to -2} \frac{1}{x^2-2x+4}$$

$$\lim_{x \to -2} \frac{1}{(-2)^2-2(-2)+4} \Rightarrow \frac{1}{4+4+4} = \frac{1}{12}$$

12. $\lim_{x \to -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4}$

$$\lim_{x \to -4} \frac{\cancel{(x-4)}(x-1)}{\cancel{(x-4)}(x+1)}$$

$$\underline{-4} + \underline{(-1)} = -5$$

$$\underline{-4} \cdot \underline{(-1)} = 4$$

$$\lim_{x \to -4} \frac{x-1}{x+4}$$

$$\underline{-4} + \underline{1} = -3$$

$$\underline{-4} \cdot \underline{(1)} = -4$$

$$\lim_{x \to -4} \frac{-4-1}{-4+1} \Rightarrow \frac{-5}{-3} = \frac{5}{3}$$

14. $\lim_{x \to -1} \frac{x^2 - 4x}{x^2 - 3x - 4}$

$$\lim_{x \to -1} \frac{\cancel{(x+4)}(x+0)}{\cancel{(x+4)}(x-1)} \quad \lim_{x \to -1} \frac{-1}{-1-1} \Rightarrow \frac{-1}{-2} = \frac{1}{2}$$

$$\underline{4} + \underline{(-1)} = 3$$

$$\underline{4} \cdot \underline{(-1)} = -4$$

$$\underline{4} + \underline{0} = 4$$

$$\underline{4} \cdot \underline{0} = 0$$

16. $\lim_{x \to -1} \dfrac{2x^2 + 3x + 1}{x^2 - 2x - 3}$

$\boxed{2x^2 + 3x + 1}$

$a = 2 \qquad \Delta = b^2 - 4ac$

$b = 3 \qquad = 3^2 - 4 \cdot 2 \cdot 1$

$c = 1 \qquad = 9 - 8$

$= 1$

$x = \dfrac{-b \pm \sqrt{\Delta}}{2 \cdot a}$

$x' = \dfrac{-3 + 1}{4} = \dfrac{-2}{4} = -\dfrac{1}{2}$

$x'' = \dfrac{-3 - 1}{4} = -\dfrac{4}{4} = -1$

$x^2 - 2x - 3$

$\dfrac{3}{1} + \dfrac{(-1)}{1} = 2$

$\dfrac{3}{1} \cdot \dfrac{(-1)}{1} = -3$

$\lim_{x \to -1} \dfrac{\left(x - \frac{1}{2}\right)\cancel{(x+1)}}{(x+3)\cancel{(x+1)}}$

$\lim_{x \to -1} \dfrac{-1 - \frac{1}{2}}{-1 + 3} = \dfrac{\frac{-3}{2}}{\frac{2}{1}} = \dfrac{-6}{2} = -3$

18. $\lim_{h \to 0} \frac{(2+h)^3 - 8}{h}$

$a^3 + 3a^2b + 3ab^2 + b^3$

$2^3 + 3(2)^2 h + 3(2)h^2 + h^3$

$8 + 12h + 6h^2 + h^3$

$\lim_{h \to 0} \frac{h^3 + 6h^2 + 12h + \cancel{8-8}}{h}$

$\lim_{h \to 0} \frac{\cancel{h}(h^2 + 6h + 12)}{\cancel{h}}$

$\lim_{h \to 0} \cancel{h^2}^{\to 0} + \cancel{6h}^{\to 0} + 12 = 12$

20. $\lim_{t \to 1} \frac{t^4 - 1}{t^3 - 1}$

$$\lim_{t \to 1} \frac{t^4 - 1}{t^3 - 1} \Rightarrow \lim_{t \to 1} \frac{(t-1)\cancel{(t+1)}(t^2+1)}{\cancel{(t+1)}(t^2-t+1)} \Rightarrow \lim_{t \to 1} \frac{\cancel{(1-1)}^{0}(\cancel{1^2+1})^{2}}{1^2-1\cancel{+1}^{0}}$$

$$= \frac{2}{1}$$

$t^3 - 1$

$(a-b)(a^2+ab+b^2)$

$(t-(-1))(t^2+t(-1)+(-1)^2)$

$(t+1)(t^2-t+1)$

$t^4 - 1$

$(t^2-1)(t^2+1)$

$(t-1)(t+1)(t^2+1)$